Evaluation of BERT-Based Models on Patient Data from French Social Media

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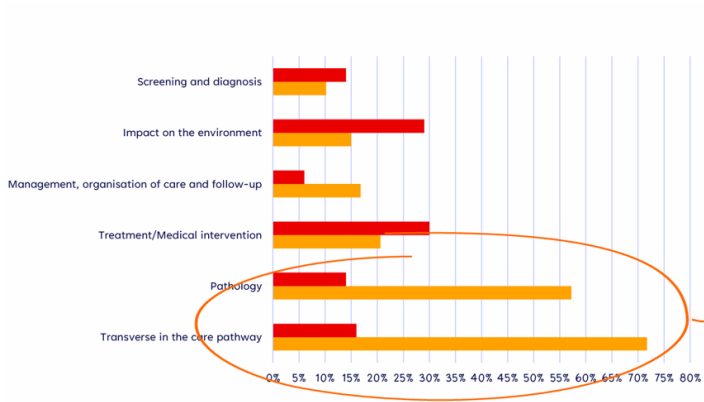








Patients express themselves differently online



- « Pathology » category refers to the expression of symptoms
- « Transverse in the care pathway » refers to taboo questions about sexuality or menstruations, anger, mental burden

In both cases, these issues are more easily discussed online than during an interview.

Mebarki, Peixoto, and El Hayani 2023

Posting about health: almost 1/5 adults and 1/3 parents

In France, Mebarki, Peixoto, and El Hayani 2023 estimated that **18% of the adults post at least occasionally online about their health or someone close's**, and 6% do so more than once a week.

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In specific subgroups:

- Men: 22%
- 18-34 yo: 31%
- Lower socio-professional category: 25%
- Long duration disease: 23%
- **Caregivers: 26%**
- **Parents of children $<$ 15yo: 30%**

KCNB1 encephalopathy: a rare and severe disease

- Developmental and Epileptic Encephalopathies (DEE): group of rare and severe diseases which commonly begin in **infancy** or **childhood** associated with frequent **seizures** of different types, **intellectual disability** and **developmental delay, plateau or regression**.
- **KCNB1 encephalopathy**:
  - only known since 2014
  - wide phenotypic spectrum
  - diverse symptoms including drug-resistant epileptic crises, spasms, severe intellectual disability, autistic disorders, developmental delays, anxiety, sleep and eating disorders, hyperactivity, ...

ASSOCIATION KCNB1 FRANCE

Leveraging social media to better understand KCNB1 encephalopathy?

Long term objective

Contributing to a better understanding of KCNB1 encepahalopathy and of its natural history using KCNB1 France's Facebook group and other online resources.

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To do so, a first step is being able to extract specific entities from this data.

- One task: Named Entity Recognition (NER) of two entities of interest - **phenotypes** (clinical signs, symptoms and conditions) and **treatments**.
- Three pre-trained models: **CamemBERT**, **CamemBERT-bio** and **DrBERT**.

⇒ **Asses the performance of these three pre-trained models on the NER task defined above on social media messages from patients and caregivers discussing health-related topics.**

Data and annotation process

- Extraction using key words on **medical forums**, public posts on **X** (Twitter) and **Facebook**.
- **16k messages about DEE** posted between 2013 and 2023[1] → partially annotated by one author in two steps: (i) regex to detect terms from the Human Phenotype Ontology, the UMLS and from a list of drugs; (ii) manual validation, addition or deletion of terms.
- **2k messages** from patients and caregivers from **broader coverage** → annotated by eight annotators ($\kappa = 0.65$).

Total number of entities annotated - IOB2 format

3,756 phenotype entities and **1,964** treatment entities

[1] extracted on a list of DEE based on Gallop et al. 2021 and Salom et al. 2023.

Three different pre-training strategies

	CamemBERT	CamemBERT-bio	DrBERT
Pre-training	MLM task with word and subword masking	MLM task with word masking	**1. full model from scratch** 2. continuing the pre-training of CamemBERT 3. continuing the pre-training of a domain specific model
Dataset (size)	OSCAR (138 GB)	biomed-fr (2.7 GB)	NACHOS (4/7.4 GB)
Version used	camembert-base	camembert-bio-base	DrBERT-7GB

Labrak et al. 2023; Martin et al. 2020; Touchent, Romary, and Clergerie 2023

Experimental set-up

- Train: 70%, validation: 15%, test: 15%
- Ten runs for each of the three pre-trained models, with the same settings.
  - 10 epochs
  - learning rate: 1e-4
  - weight decay: 0.01
- The **precision**, **recall** and **weighted F1-score** were calculated using the same framework as in Touchent, Romary, and Clergerie 2023: average of the scores on 10 runs, using seqeval on strict mode[2].
- The weighted F1-score on our "social media" task was compared to the ones on **MEDLINE** and **EMEA** tasks, given in ibid.

[2] i.e. an entity is considered correctly predicted if both its type and boundaries exactly match the reference.

Comparison of the fine-tuned models on the Social Media Testimonies NER task

		CamemBERT	CamemBERT-bio	DrBERT
phenotype	f1-score	0.55	**0.56**	0.49
	precision	**0.62**	0.58	0.58
	recall	0.50	**0.53**	0.43
	support	*610*	*610*	*513*
treatment	f1-score	0.64	**0.65**	0.64
	precision	0.58	0.56	**0.62**
	recall	0.72	**0.78**	0.65
	support	*340*	*340*	*278*
weighted-average	f1-score	0.58	**0.59**	0.54
	precision	**0.61**	0.57	0.59
	recall	0.58	**0.62**	0.51
	support	*950*	*950*	*791*

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Comparison of weighted F1-scores on three different NER tasks

	CamemBERT	CamemBERT-bio	DrBERT
EMEA task	0.88	0.90	0.87
MEDLINE task	0.76	0.78	0.76
"Social media" task	0.58	0.59	0.54

Touchent, Romary, and Clergerie 2023

Leveraging social media testimonies for health: a new challenge?

- Close performances of CamemBERT and CamemBERT-bio $\rightarrow$ *on social media data, the language pre-training seems to be more relevant than the domain pre-training.*

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- All three models performed better on more "traditional" datasets → *future work to find a strategy better suited to this specific kind of data*.
- Limitations : the dataset was not annotated using a single process and with low inter-annotator agreement + task that could be better defined, more precise.

Future work

- **New annotation** with more precise entities and a more systematic approach.
- Exploring fine-tuning of **other pre-trained models** such as SocBERT (Guo and Sarker 2023), BERTweet (D. Q. Nguyen, Vu, and A. T. Nguyen 2020), ...
- Exploring **generative models**, combinations of different approaches.
- **Applying the new pipeline to posts from KCNB1 France's Facebook group** to try to understand better KCNB1 encephalopathy.

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Salom, Rafael et al. (Aug. 2023). "Dataset on the psychosocial impact in families with children with developmental and epileptic encephalopathies". In: *Scientific Data* 10, p. 530. DOI: 10.1038/s41597-023-02441-3.

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Thank you!



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